Vector at Pos "t"

$$\begin{bmatrix} \sin(\omega t) \\ \cos(\omega t) \end{bmatrix} \rightarrow \begin{bmatrix} \sin(a) \\ \cos(a) \end{bmatrix}$$

Vector at Pos "t+δ"

$$\begin{bmatrix} \text{Sin}\,(\omega(t+\delta)) \\ \cos\,(\omega(t+\delta)) \end{bmatrix} = \begin{bmatrix} \sin(\omega t)\cos(\omega\delta) + \cos(\omega t)\sin(\omega\delta) \\ \cos(\omega t)\cos(\omega\delta) - \sin(\omega t)\sin(\omega\delta) \end{bmatrix} \rightarrow \begin{bmatrix} \sin(a)\cos(b) + \cos(a)\sin(b) \\ \cos(a)\cos(b) - \sin(a)\sin(b) \end{bmatrix}$$

Let $\omega t = a$, $\omega\delta = b$

$\sin^2(a) + \cos^2(a) = 1$

Squared Distance Between Vector of Pos "t+δ" and Vector of Pos "t" [Two Consecutive Positions]

$$[\sin(a)\cos(b) + \cos(a)\sin(b) - \sin(a)]^2 + [-\sin(a)\sin(b) + \cos(a)\cos(b) - \cos(a)]^2 =$$

$$[\sin(a)(\cos(b)-1) + \cos(a)\sin(b)]^2 + [-\sin(a)\sin(b) + \cos(a)(\cos(b)-1)]^2 =$$

$$[\sin(a)^2(\cos(b)-1)^2 + \cos^2(a)\sin^2(b) + 2\sin(a)\cos(a)\sin(b)(\cos(b)-1)] +$$
$$[\sin^2(a)\sin^2(b) + \cos^2(a)(\cos(b)-1)^2 - 2\sin(a)\cos(a)\sin(b)(\cos(b)-1)] =$$

$$\sin(a)^2(\cos(b)-1)^2 + \cos^2(a)\sin^2(b) + \sin^2(a)\sin^2(b) + \cos^2(a)(\cos(b)-1)^2 =$$

$$[\sin(a)^2 + \cos^2(a)](\cos(b)-1)^2 + [\sin(a)^2 + \cos^2(a)](\sin^2(b)) = (\cos(b)-1)^2 + \sin^2(b) =$$

$$\cos^2(b) - 2\cos(b) + 1 + \sin^2(b) = 2 - 2\cos(b) = 2(1-\cos(b))$$

$$= \mathbf{2(1 - \cos(\omega\delta))}$$

Irrespective of Position "t"